

EDUCATION

National University of Computer and Emerging Sciences
Bachelors in Artificial Intelligence (2021 – Ongoing)

Islamabad, Pakistan

EMPLOYMENT

Software Developer

IVY Interactive Solutions, Islamabad, Pakistan

May 2025 – Present

- Automated project planning, reducing manual effort and accelerating delivery.
- Developed a dashboard for real-time analytics, enabling early risk detection.
- Collaborate with cross-functional teams to deliver scalable software solutions.
- Support timely delivery by maintaining technical documentation and assisting with deployments.

AI Intern

CarbonTeq, Lahore, Pakistan

July 2024 – September 2024

- Delivered a Retrieval-Augmented Generation (RAG) system, improving AI response relevance.
- Enhanced business insights by deploying ML models for predictive analytics.
- Automated model deployment with CI/CD, reducing release time and errors.

Full Stack Development Intern

CarbonTeq, Lahore, Pakistan

July 2022 – September 2022

- Built a DNS client, streamlining hostname-to-IP resolution for internal tools.
- Secured data flow by developing RESTful APIs with JWT authentication and MongoDB.
- Worked in Agile teams, collaborating on daily sprints and stand-ups.

TECHNICAL SKILLS

AI/ML: Python, TensorFlow, PyTorch, LangChain, HuggingFace, OpenAI API, Scikit-learn

NLP/Generative AI: RAG, LLMs (GPT, LLaMA), TTS, Transformer models, Prompt Engineering

Backend: Node.js, Flask, Express, REST, GraphQL, Docker, CI/CD

Databases: MongoDB, PostgreSQL, Firebase, SQL

Frontend: React, Next.js, Svelte, Tailwind, JS, HTML, CSS

Monitoring: Prometheus, Grafana, MLflow

PROJECTS

RAG-Based Chatbot System

Retrieval-Augmented Generation (RAG)

- Built a RAG chatbot for document-based Q&A, improving information retrieval and response quality.
- Integrated Gemini 1.5 flash to boost inference speed and answer accuracy.

LLM-Podcast-Generator

AI-Driven Content Generation

- Automated podcast script generation from research papers, reducing manual content creation effort.
- Improved script quality by combining multiple LLM approaches.
- Enabled seamless podcast audio conversion with TTS integration.

CycleGAN on Person Sketches

Image-to-Image Translation

- Implemented CycleGAN for sketch-to-portrait translation, achieving high-quality bidirectional results.

- Improved visual outputs using cycle-consistency and adversarial losses.
- Enhanced model interpretability with training visualizations.

Trend2Story

Real-Time Trend Analysis & Automated Storytelling

- Automated trending topic extraction and creative story generation, streamlining content workflows.
- Delivered user-selectable story themes for greater engagement.
- Simplified deployment and maintenance with containerization.

Multi-Modal Sentiment Analysis

Deep Learning

- Combined text and image data for sentiment classification, increasing model accuracy.
- Improved sentiment prediction by fusing visual and textual features.